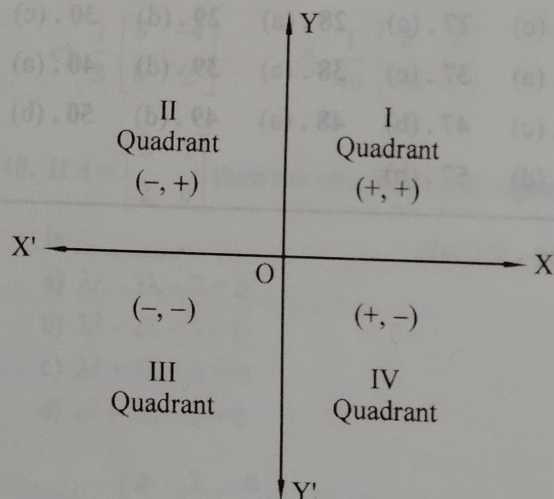


Straight Lines

2.1 CO-ORDINATE SYSTEM

Analytical geometry deals with the study of the geometrical properties of points and lines by the use of algebraic methods. Thus, it is called algebraic geometry or co-ordinate geometry.

The position of an object or a point is located using cartesian plane



2.2 SOME IMPORTANT FORMULA

1. Distance of a point $P(x, y)$ from origin $O(0, 0)$

$$D = \sqrt{x^2 + y^2}$$

2. Distance between the points (x_1, y_1) and (x_2, y_2)

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

3. Midpoint of the line, $M = \left[\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right]$

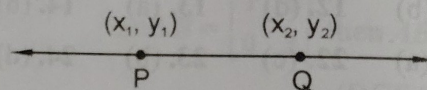
4. Centroid of the triangle

$$C = \left[\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right]$$

2.3 STRAIGHT LINES

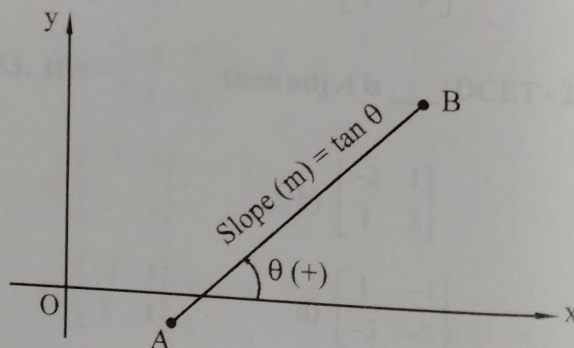
The straight line is the locus of point which moves without changing its direction

A straight line is a figure formed when two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ are connected between them, and the line ends are extended to infinity.



Slope / Gradient of a straight line:

If a straight line makes an angle θ with the positive direction of the x -axis, then θ is called the inclination and tangent angle ($\tan \theta$) is called the slope (m) or gradient of the line.



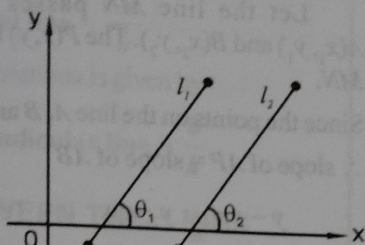
The slope of a line is generally given as $m = \tan \theta$

Note :

1. Slope is **positive**, when θ is an acute angle i.e., θ lies between 0° and 90° .
2. Slope is **negative**, when θ is an obtuse angle i.e., θ lies between 90° and 180° .
3. Slope is **zero**, when θ is equal to 0° or it is **parallel to x -axis**.
4. Slope is **infinite**, when θ is equal to 90° or it is **perpendicular to x -axis**.

2.4 SLOPES OF PARALLEL LINES

Let l_1 and l_2 be two non-vertical parallel lines with respective inclination θ_1 and θ_2 with the positive direction of the x-axis.



Here l_1 and l_2 be parallel lines.

Hence their inclinations of the lines are equal.

$$\therefore \theta_1 = \theta_2$$

$$\text{then } \tan \theta_1 = \tan \theta_2$$

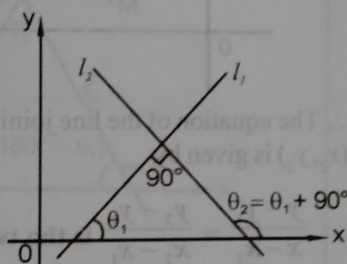
$$\therefore m_1 = m_2$$

$$\therefore \tan \theta = m$$

Hence slope of two parallel lines are equal.

2.5 SLOPE OF PERPENDICULAR LINES

Let l_1 and l_2 be two non-vertical lines with respective inclination θ_1 and θ_2 with the positive direction of the x-axis.



Here l_1 and l_2 be perpendicular lines. Hence inclination of the line l_2 is

$$\theta_2 = \theta_1 + 90^\circ$$

$$\text{then } \tan \theta_2 = \tan (\theta_1 + 90^\circ)$$

$$\tan \theta_2 = -\cot \theta_1$$

$$\tan \theta_2 = -\frac{1}{\tan \theta_1}$$

$$\tan \theta_1 \cdot \tan \theta_2 = -1$$

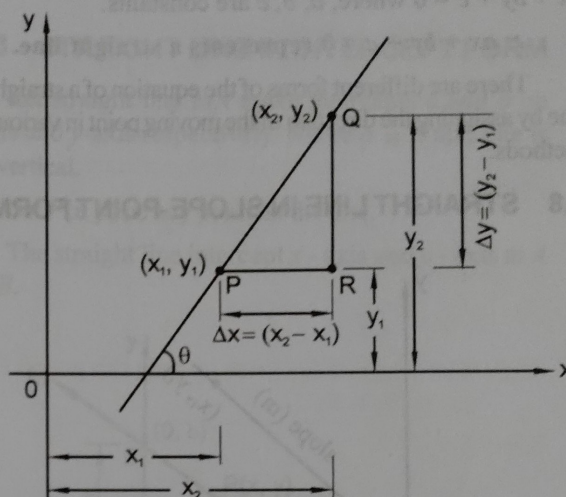
$$\therefore m_1 \cdot m_2 = -1$$

Hence slope of two perpendicular lines are product of slope of l_1 and l_2 be minus one.

2.6 SLOPE OF LINE JOINING TWO POINTS

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be the given points on the line and θ be the inclination of the line.

The slope of a line is the ratio of change in Y (i.e., Δy) to the change in X (i.e., Δx) between two distinct points on a line.



Generally, the slope of a line is given as $m = \tan \theta$.

$$\therefore m = \tan \theta = \frac{\text{Change in } Y \text{ (i.e., } \Delta y)}{\text{Change in } X \text{ (i.e., } \Delta x)}$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$\therefore$ The slope of the line joining two points

$$(x_1, y_1) \text{ and } (x_2, y_2), \text{ i.e., } m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Illustration,

If $P = (3, 2)$ and $Q = (5, -4)$ are the points on a line, then the slope of line is,

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4 - 2}{5 - 3} = \frac{-6}{2} = -3$$

Slope of line having different inclination with x-axis

(Inclination) Angle (θ)	0°	30°	45°	60°	90°	120°	135°	150°	180°
Slope (m)	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

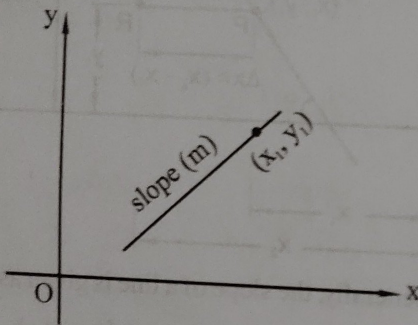
2.7 DESCRIPTION OF A LINE BY AN EQUATION

A first degree equation in x, y i.e., $ax + by + c = 0$ represents a line, it means that all points (x, y) satisfying $ax + by + c = 0$ lies along a line. Thus straight line can also defined as the locus of a point satisfying the condition $ax + by + c = 0$ where, a, b, c are constants.

$\therefore ax + by + c = 0$ represents a straight line.

There are different forms of the equation of a straight line by assigning the direction of the moving point in various methods.

2.8 STRAIGHT LINE IN SLOPE-POINT FORM

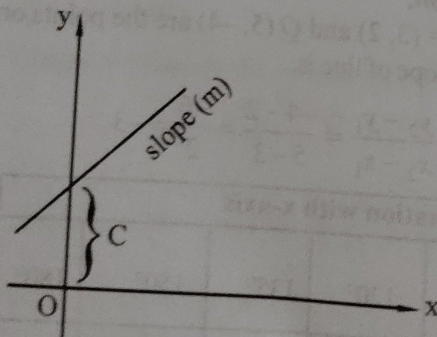


The slope of line, $m = \tan \theta$ and point (x_1, y_1) of equation of a line is $(y - y_1) = m(x - x_1)$

$\therefore y - y_1 = m(x - x_1)$ is the slope point form

2.9 STRAIGHT LINE IN SLOPE - INTERCEPT FORM

Slope (m) and y -intercept (c) form of equation of a line is $y = mx + c$



$\therefore y = mx + C$ is the slope-intercept form

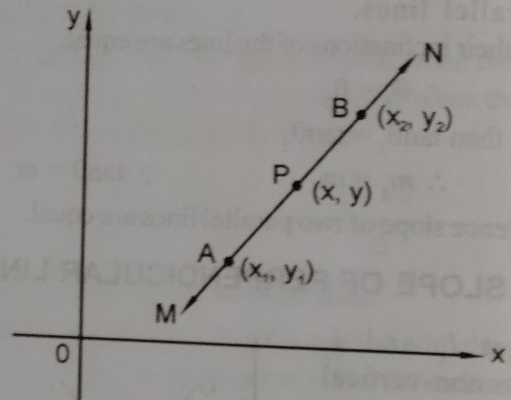
2.10 STRAIGHT LINE IN TWO - POINT FORM

Let the line MN passes through the points $A(x_1, y_1)$ and $B(x_2, y_2)$. The $P(x, y)$ be any point on the line MN .

Since the points on the line A, B and P are collinear

$\therefore$ slope of AP = slope of AB

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$



$\therefore$ The equation of the line joining the points (x_1, y_1) and (x_2, y_2) is given by

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1} \text{ is the two point form}$$

2.11 STRAIGHT LINE IN GENERAL FORM AND INTERCEPTS

The general equation of a straight line, $ax + by + c = 0$ where a, b and c are constants.

If $a \neq 0, b \neq 0, c \neq 0$ then

a) Slope of a line is given by

$$\therefore \text{slope } m = \frac{-a}{b} = \frac{-\text{coefficient of } x}{\text{coefficient of } y}$$

b) x -intercept of the line is given by,

$$\therefore x\text{-intercept} = \frac{-c}{a} = \frac{-\text{constant term}}{\text{coefficient of } x}$$

c) y -intercept of the line is given by,

$$\therefore y\text{-intercept} = \frac{-c}{b} = \frac{-\text{constant term}}{\text{coefficient of } y}$$

d) Slope of parallel line is given by

$$\therefore \text{slope of parallel line} = \frac{-a}{b}$$

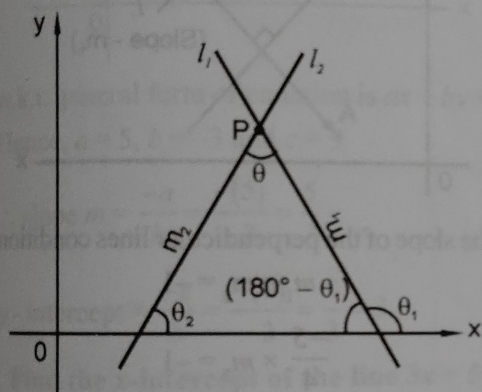
e) Slope of perpendicular line is given by

$$\therefore \text{Slope of perpendicular line} = \frac{b}{a}$$

2.12 ANGLE BETWEEN TWO LINES

Let l_1 and l_2 be the two non vertical lines intercept at point P having a inclination θ_1 and θ_2 with x-axis respectively.

The slope of lines l_1 and l_2 are $m_1 = \tan\theta_1$ and $m_2 = \tan\theta_2$ respectively.



We know that, The sum of interior angles of a triangle is equal to 180°

$$\text{i.e., } \theta + \theta_2 + (180^\circ - \theta_1) = 180^\circ$$

$$\theta + \theta_2 - \theta_1 = 0$$

$$\therefore \theta = \theta_1 - \theta_2$$

$$\therefore \tan\theta = \tan(\theta_1 - \theta_2)$$

$$\text{i.e., } \tan\theta = \frac{\tan\theta_1 - \tan\theta_2}{1 + \tan\theta_1 \cdot \tan\theta_2}$$

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$\text{or } \theta = \tan^{-1} \left[\frac{m_1 - m_2}{1 + m_1 m_2} \right]$$

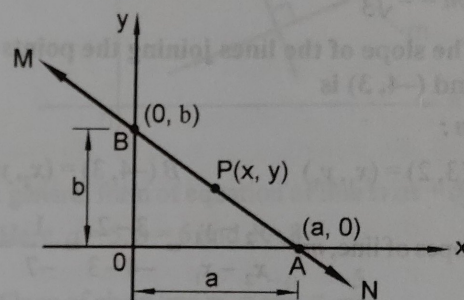
$\therefore$ The angle between the lines whose slope m_1 and m_2 is given by $\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2}$

2.13 STRAIGHT LINE IN INTERCEPT FORM

Let straight line MN make intercepts a and b on x-axis and y-axis respectively, where $a \neq 0$ and line is non vertical.

Let $P(x, y)$ be any point on the line.

The straight line intercept x - axis and y - axis at A and B .



Since the line passes through A, B and P ,

$\therefore$ slope of AB = slope of AP

$$\frac{b - 0}{0 - a} = \frac{y - 0}{x - a}$$

$$b(x - a) = -a(y - 0)$$

$$bx - ab = -ay$$

$$bx + ay = ab \text{ ---- divide by } ab$$

$$\frac{bx}{ab} + \frac{ay}{ab} = \frac{ab}{ab}$$

$$\therefore \frac{x}{a} + \frac{y}{b} = 1 \text{ is the intercept form of required line.}$$

EXAMPLE PROBLEMS

1. What is the slope of a line whose inclination is 45° .

Solution : Given $\theta = 45^\circ$
 $\therefore$ slope, $m = \tan\theta = \tan 45^\circ$
 $m = 1$

2. The slope of the line making an angle 120° with x-axis is

Solution :
 Slope of the line, $m = \tan\theta = \tan 120^\circ$
 i.e., $m = \tan(180^\circ - 60^\circ) \Rightarrow m = -\tan 60^\circ$
 $\therefore m = -\sqrt{3}$

3. The slope of the lines joining the points (3, 2) and (-4, 3) is

Solution :
 $A(3, 2) = (x_1, y_1)$ $B(-4, 3) = (x_2, y_2)$
 Slopes of line, $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 2}{-4 - 3} = \frac{1}{-7} = -\frac{1}{7}$

4. Find the slope of the line joining the points (2, -3) (6, 1).

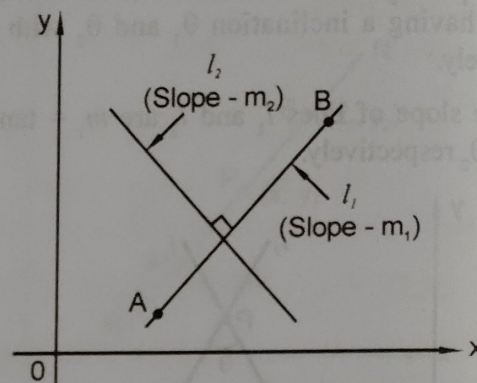
Solution : Given $P(x_1, y_1) = (2, -3)$ and $Q(x_2, y_2) = (6, 1)$
 The slope of line, $m = \frac{y_2 - y_1}{x_2 - x_1}$
 $m = \frac{1 - (-3)}{6 - 2} = \frac{1 + 3}{4} = \frac{4}{4}$
 $\therefore m = 1$

5. Find the slope of the line parallel to the line joining the points (2, 3) (4, -1).

Solution : Given $A(x_1, y_1) = (2, 3)$ and $B(x_2, y_2) = (4, -1)$
 Slope of line, $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1 - 3}{4 - 2} = \frac{-4}{2} = -2$
 w.k.t slope of the parallel lines are equal.
 $m_1 = m_2 = m = -2$
 $\therefore$ Slope of parallel line is -2

6. Find the slope of the line perpendicular to the line joining the points (-3, 6) and (5, -4).

Solution : Given $A(x_1, y_1) = (-3, 6)$ and $B(x_2, y_2) = (5, -4)$
 Slope of line, $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4 - 6}{5 + 3} = \frac{-10}{8} = -\frac{5}{4}$



The slope of the perpendicular lines condition is

$$m_1 \times m_2 = -1$$

$$-\frac{5}{4} \times m_2 = -1$$

$$m_2 = \frac{-1(4)}{-5} = \frac{-4}{-5} = \frac{4}{5}$$

$\therefore$ Slope of perpendicular line is $\frac{4}{5}$

7. If the slope of the line joining the points (2, 5) and (3, λ) is -2 , find the value of λ .

Solution : Given $A(x_1, y_1) = (2, 5)$, $B(x_2, y_2) = (3, \lambda)$ and $m = -2$

$$\text{Slope of line, } m = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow -2 = \frac{(\lambda - 5)}{(3 - 2)}$$

$$-2(3 - 2) = \lambda - 5$$

$$-6 + 4 = \lambda - 5$$

$$-2 + 5 = \lambda \quad \therefore \lambda = 3$$

8. Find the slope of the line $3x - 4y + 8 = 0$.

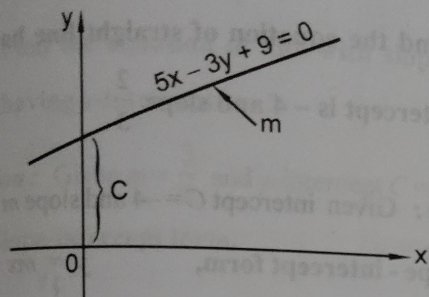
Solution : Given $3x - 4y + 8 = 0$

w.k.t. general form of equation is $ax + by + c = 0$
Hence, $a = 3$, $b = -4$ and $c = 8$

$$\therefore \text{slope } m = \frac{-a}{b} = \frac{-(3)}{-4} = \frac{3}{4}$$

9. Find the slope and y-intercept of the straight line $5x - 3y + 9 = 0$.

Solution : Given $5x - 3y + 9 = 0$



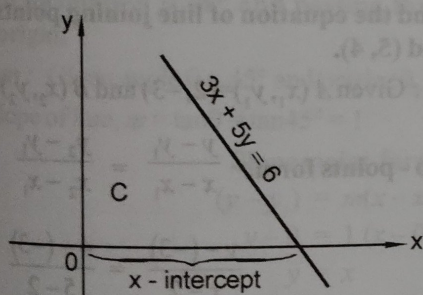
w.k.t. general form of equation is $ax + by + c = 0$
Hence, $a = 5$, $b = -3$ and $c = 9$

$$\therefore \text{slope } m = \frac{-a}{b} = \frac{-(5)}{-3} = \frac{5}{3}$$

$$y\text{-intercept} = \frac{-c}{b} = \frac{-(9)}{-3} = \frac{9}{3} = 3$$

10. Find the x-intercept of the line $3x + 5y = 6$.

Solution : Given $3x + 5y = 6 \Rightarrow 3x + 5y - 6 = 0$



w.k.t. general form of equation is $ax + by + c = 0$
Hence, $a = 3$, $b = 5$ and $c = -6$

$$\therefore x\text{-intercept} = \frac{-c}{a} = \frac{-(-6)}{3} = \frac{6}{3} = 2$$

11. Find the slope of line parallel to $3x - 2y + 2 = 0$.

Solution : Given $3x - 2y + 2 = 0$

w.k.t. general form of equation is $ax + by + c = 0$

Here, $a = 3$, $b = -2$ and $c = 2$

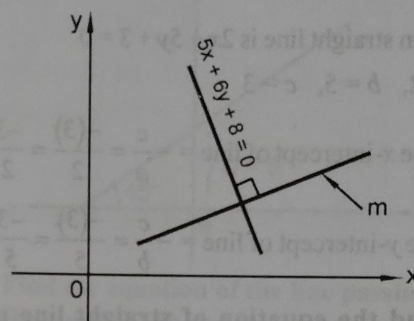
$$\text{Slope of line } m = \frac{-a}{b} = \frac{-3}{-2} = \frac{3}{2}$$

and slope of parallel lines are same (i.e., $m_1 = m_2$)

$$\therefore \text{slope of parallel line is } \frac{3}{2}$$

12. Find the slope of line perpendicular to $5x + 6y + 8 = 0$.

Solution : Given $5x + 6y + 8 = 0$



w.k.t. general form of equation of line is $ax + by + c = 0$

Here, $a = 5$, $b = 6$ and $c = 8$

$$\text{Slope of given line } m_1 = \frac{-a}{b} = \frac{-5}{6}$$

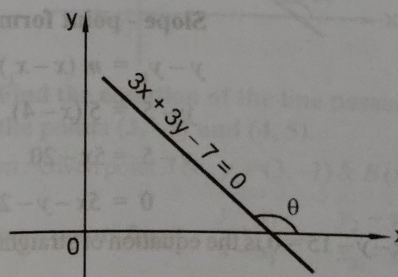
and slope of perpendicular lines condition,
i.e., $m_1 \times m_2 = -1$

$$\frac{-5}{6} \times m_2 = -1 \Rightarrow m_2 = \frac{-1 \times 6}{-5} = \frac{6}{5}$$

$$\therefore \text{slope of perpendicular line is } \frac{6}{5}$$

13. Find the inclination of line $3x + 3y - 7 = 0$ with x-axis.

Solution : Given $3x + 3y - 7 = 0$



w.k.t. general form of equation of line is $ax + by + c = 0$

Here, $a = 3$, $b = 3$ and $c = -7$

$$\text{Slope } m = \frac{-a}{b} = \frac{-3}{3} = -1$$

$$\text{and } \tan \theta = m$$

$$\text{w.k.t. } \tan 135^\circ = -1$$

$$\therefore \theta = 135^\circ$$

14. The x-intercept and y-intercept of the straight line $2x + 5y + 3 = 0$ is

Solution :

Given straight line is $2x + 5y + 3 = 0$

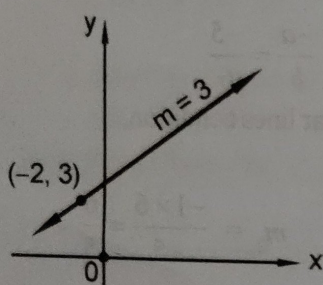
$$a = 2, \quad b = 5, \quad c = 3$$

$$\therefore \text{the x-intercept of line} = -\frac{c}{a} = \frac{-(-3)}{2} = \frac{-3}{2}$$

$$\text{the y-intercept of line} = -\frac{c}{b} = \frac{-(-3)}{5} = \frac{-3}{5}$$

15. Find the equation of straight line passing through the point $(-2, 3)$ & having slope 3.

Solution : Given, point $(x_1, y_1) = (-2, 3)$ and slope $m = 3$



Slope - point form

$$y - y_1 = m(x - x_1)$$

$$y - 3 = 3[x - (-2)]$$

$$y - 3 = 3(x + 2)$$

$$y - 3 = 3x + 6$$

$$0 = 3x - y + 6 + 3$$

$\therefore 3x - y + 9 = 0$ is the equation of straight line.

16. Find the equation of line passing through the point $(4, 5)$ and having slope 5.

Solution : Given, point $(x_1, y_1) = (4, 5)$ and slope $m = 5$

Slope - point form

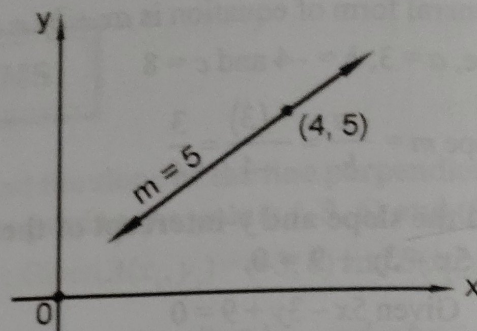
$$y - y_1 = m(x - x_1)$$

$$y - 5 = 5(x - 4)$$

$$y - 5 = 5x - 20$$

$$0 = 5x - y - 20 + 5$$

$\therefore 5x - y - 15 = 0$ is the equation of straight line

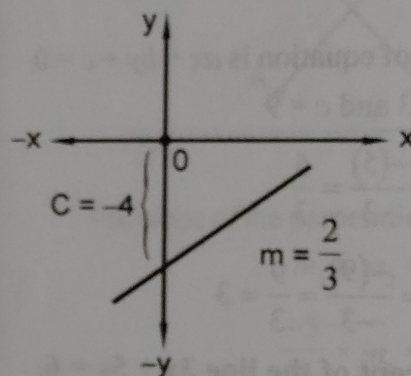


17. Find the equation of straight line having y-intercept is -4 and slope $\frac{2}{3}$

Solution : Given intercept $C = -4$ and slope $m = \frac{2}{3}$

Slope - intercept form,

$$y = mx + C$$



$$y = \frac{2}{3}x + (-4)$$

$$y + 4 = \frac{2}{3}x$$

$$(y + 4)3 = 2x$$

$$3y + 12 = 2x$$

$2x - 3y - 12 = 0$ is the equation of line.

18. Find the equation of line joining points $(2, -3)$ and $(5, 4)$.

Solution : Given $A(x_1, y_1) = (2, -3)$ and $B(x_2, y_2) = (5, 4)$

Two - points form, $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$

$$\frac{y - (-3)}{x - 2} = \frac{4 - (-3)}{5 - 2}$$

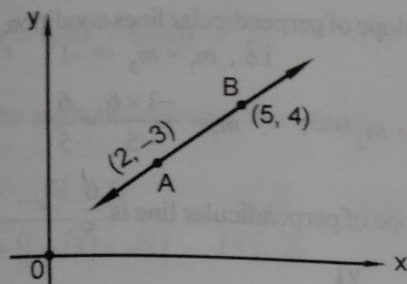
$$y + 3 = \frac{7}{3}(x - 2)$$

$$(y + 3)3 = 7(x - 2)$$

$$3y + 9 = 7x - 14$$

$$0 = 7x - 3y - 14 - 9$$

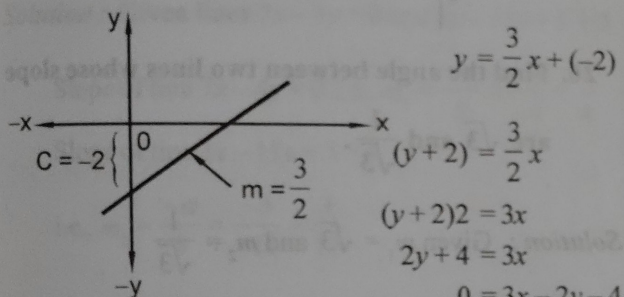
$7x - 3y - 23 = 0$ is the equation of line.



19. Find the equation of line with slope $\frac{3}{2}$ and having y-intercept is -2.

Solution : Given $m = \frac{3}{2}$ and y-intercept $C = -2$

Slope-intercept form,



$\therefore 3x - 2y - 4 = 0$ is the equation of line.

20. Find the equation of straight line whose angle of inclination is 45° and passing through the origin.

Solution : Given angle $\theta = 45^\circ$ and origin $(0, 0)$.

Slope of line, $m = \tan\theta = \tan 45^\circ = 1$

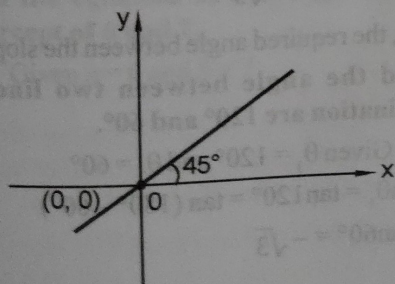
Slope-point form

$$(y - y_1) = m(x - x_1)$$

$$y - 0 = 1(x - 0)$$

$$y = x$$

$\therefore x - y = 0$ is the equation of line.



21. Find the equation of the line whose y-intercept is 5 and makes 45° with positive x-axis.

Solution : Given y-intercept $C = 5$ and $\theta = 45^\circ$

Slope of line, $m = \tan\theta = \tan 45^\circ = 1$

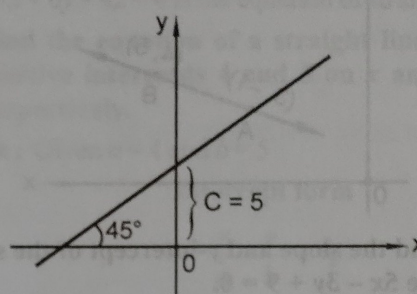
$\therefore$ **Slope - intercept form**

$$y = mx + C$$

$$y = 1x + 5$$

$$0 = x - y + 5$$

$\therefore x - y + 5 = 0$ is the equation of line.



22. Find the equation of the line passing through the point $(2, -3)$ and makes an angle 135° with the positive direction of x-axis.

Solution : Given point $(x_1, y_1) = (2, -3)$ and angle $\theta = 135^\circ$

The slope of a line $m = \tan\theta = \tan 135^\circ = \tan(90^\circ + 45^\circ)$

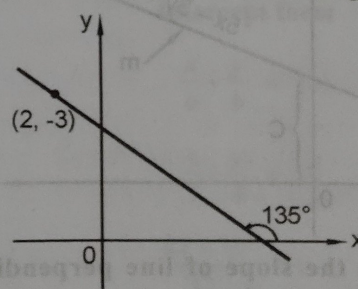
$$\therefore m = -\cot 45^\circ = -1$$

Slope-point form, $y - y_1 = m(x - x_1)$

$$y - (-3) = -1(x - 2)$$

$$y + 3 = -x + 2 \Rightarrow x + y + 3 - 2 = 0$$

$\therefore x + y + 1 = 0$ is the equation of straight line.



23. Find the equation of the line passing through the points $(3, -1)$ and $(4, 5)$.

Solution : Given point $A(x_1, y_1) = (3, -1)$ & $B(x_2, y_2) = (4, 5)$

Two - points form, $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$

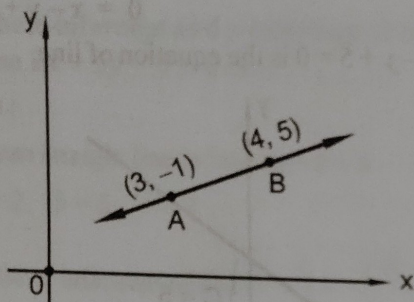
$$\frac{y - (-1)}{x - 3} = \frac{5 - (-1)}{4 - 3}$$

$$y + 1 = \frac{6}{1}(x - 3)$$

$$y + 1 = 6x - 18$$

$$0 = 6x - y - 18 - 1$$

$6x - y - 19 = 0$ is the equation of line.



24. Find the slope and y-intercept of the straight line $5x - 3y + 9 = 0$.

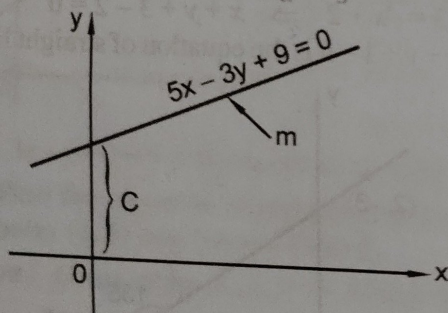
Solution : Given $5x - 3y + 9 = 0$

w.k.t. general form of equation is $ax + by + c = 0$

Hence, $a = 5$, $b = -3$ and $c = 9$

$$\therefore \text{slope } m = \frac{-a}{b} = \frac{-(5)}{-3} = \frac{5}{3}$$

$$y\text{-intercept} = \frac{-c}{b} = \frac{-(9)}{-3} = \frac{9}{3} = 3$$



25. Find the slope of line perpendicular to $5x + 6y + 8 = 0$.

Solution : Given $5x + 6y + 8 = 0$

w.k.t. general form of equation of line is $ax + by + c = 0$

Here, $a = 5$, $b = 6$ and $c = 8$

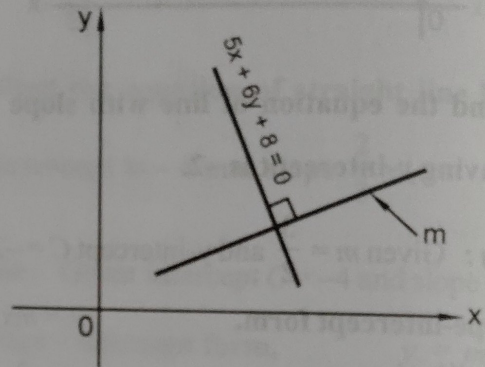
$$\text{Slope of given line } m_1 = \frac{-a}{b} = \frac{-5}{6}$$

and slope of perpendicular lines condition,

$$\text{i.e., } m_1 \times m_2 = -1$$

$$\frac{-5}{6} \times m_2 = -1 \quad ; \quad m_2 = \frac{-1 \times 6}{-5} = \frac{6}{5}$$

$\therefore$ slope of perpendicular line is $\frac{6}{5}$



26. Find the angle between two lines whose slope

are $\sqrt{3}$ and $\frac{1}{\sqrt{3}}$.

Solution : Given $m_1 = \sqrt{3}$ and $m_2 = \frac{1}{\sqrt{3}}$

The angle between the lines, $\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$

$$\tan \theta = \frac{\sqrt{3} - \frac{1}{\sqrt{3}}}{1 + \left(\sqrt{3} \times \frac{1}{\sqrt{3}}\right)} = \frac{\frac{3-1}{\sqrt{3}}}{1+1} = \frac{\left(\frac{2}{\sqrt{3}}\right)}{2}$$

$$\tan \theta = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\text{w.k.t. } \tan 30^\circ = \frac{1}{\sqrt{3}} \quad \therefore \theta = 30^\circ$$

Therefore, the required angle between the slopes is 30° .

27. Find the angle between two lines whose inclination are 120° and 60° .

Solution : Given $\theta_1 = 120^\circ$ and $\theta_2 = 60^\circ$

$$\therefore m_1 = \tan \theta_1 = \tan 120^\circ = \tan (180^\circ - 60^\circ)$$

$$= -\tan 60^\circ = -\sqrt{3}$$

$$m_2 = \tan \theta_2 = \tan 60^\circ = \sqrt{3}$$

$\therefore$ The angle between the lines, $\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$

$$\tan \theta = \frac{-\sqrt{3} - \sqrt{3}}{1 + ((-\sqrt{3}) \times \sqrt{3})} = \frac{-2\sqrt{3}}{1 + (-3)} = \frac{-2\sqrt{3}}{-2} = \sqrt{3}$$

$$\tan \theta = \sqrt{3}$$

$$\text{w.k.t. } \tan 60^\circ = \sqrt{3} \quad \therefore \theta = 60^\circ$$

Therefore, the required acute angle between the given slopes is 60° .

28. Find the acute angle between the lines $7x - 4y = 0$ and $3x - 11y + 5 = 0$.

Solution : Given lines $7x - 4y = 0$ and $3x - 11y + 5 = 0$

$$\text{Slope of line } 7x - 4y = 0 \text{ i.e., } m_1 = \frac{-a}{b} = \frac{-7}{-4} = \frac{7}{4}$$

$$\text{Slope of line } 3x - 11y + 5 = 0$$

$$\text{i.e., } m_2 = \frac{-a}{b} = \frac{-3}{-11} = \frac{3}{11}$$

The angle between the lines, $\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$

$$= \frac{\frac{7}{4} - \frac{3}{11}}{1 + \left(\frac{7}{4} \times \frac{3}{11}\right)} = \frac{\left(\frac{77 - 12}{44}\right)}{\left(1 + \frac{21}{44}\right)} = \frac{\left(\frac{65}{44}\right)}{\left(\frac{44 + 21}{44}\right)} = \frac{\left(\frac{65}{44}\right)}{\left(\frac{65}{44}\right)}$$

$$\tan \theta = 1$$

$$\text{w.k.t. } \tan 45^\circ = 1 \quad \therefore \theta = 45^\circ$$

Therefore, the required acute angle between the given lines is 45° .

29. Find the equation of a straight line having intercept of 6 and 7.

Solution : Given $a = 6$ and $b = 7$

Intercept form

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{x}{6} + \frac{y}{7} = 1$$

$$\frac{7x + 6y}{42} = 1$$

$$7x + 6y = 42$$

$\therefore 7x + 6y - 42 = 0$ is the equation of straight line.

30. Find the equation of a straight line having positive intercepts 4 and 5 on x and y axis respectively.

Solution : Given $a = 4$ and $b = 5$

Intercept form

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{x}{4} + \frac{y}{5} = 1$$

$$\frac{5x + 4y}{20} = 1$$

$$5x + 4y = 20$$

$\therefore 5x + 4y - 20 = 0$ is the equation of straight line.

31. Find the equation of a straight line having intercepts of 3 and 4.

Solution : Given $a = 3$ and $b = 4$

Intercept form

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{x}{3} + \frac{y}{4} = 1$$

$$\frac{4x + 3y}{12} = 1$$

$$4x + 3y = 12$$

$\therefore 4x + 3y - 12 = 0$ is the equation of straight line.

PRACTICE MCQ's

1. The slope of line joining the points (3, -2) and (5, 4) is

- a) $\frac{1}{3}$ b) $\frac{2}{3}$ c) 3 d) $\frac{3}{2}$

2. The slope of the line making an angle 135° with the x-axis is

- a) $-\sqrt{3}$ b) 1
c) -1 d) $\sqrt{3}$

3. The slope of straight line $2y - 3x + 7 = 0$ is

- a) $\frac{3}{2}$ b) $\frac{2}{3}$
c) $-\frac{2}{3}$ d) $-\frac{3}{2}$

4. The slope and y-intercept of the straight line $5x - 3y + 9 = 0$ is

- a) $\left(\frac{5}{3}, 3\right)$ b) $\left(\frac{3}{5}, \frac{5}{3}\right)$

- c) $\left(3, \frac{5}{3}\right)$ d) $\left(3, \frac{3}{5}\right)$

5. The x-intercept of the line $3x + 5y = 6$ is

- a) -2 b) 2
c) 3 d) 5

6. The inclination of line $2x - 2y + 5 = 0$ is

- a) 45° b) 135°
c) 30° d) 60°

7. The slope of line parallel to $3x - 4y + 8 = 0$ is

- a) $-\frac{4}{3}$ b) $\frac{4}{3}$ c) $-\frac{3}{4}$ d) $\frac{3}{4}$

8. The slope of line perpendicular to $5x + 6y + 8 = 0$ is,

- a) $-\frac{5}{6}$ b) $-\frac{6}{5}$
c) $\frac{6}{5}$ d) $\frac{5}{6}$

9. The slope of the line perpendicular to the line joining the points (-3, 6) and (5, 5) is

- a) -8 b) $\frac{1}{8}$ c) $-\frac{1}{8}$ d) 8

10. The slope of a line parallel to x-axis is

- a) 1 b) 0
c) -1 d) ∞

11. The slope of a line perpendicular to x-axis is

- a) -1 b) 0
c) ∞ d) 1

12. The equation of a line passing through (2, 3) and having slope 5 is

- a) $5x + y - 7 = 0$ b) $5x - y + 7 = 0$
c) $x - 5y + 7 = 0$ d) $5x - y - 7 = 0$

13. The equation of straight line inclined at 135° and passing through the point (1, 2) is

- a) $x - y + 3 = 0$ b) $x + y - 3 = 0$
c) $x - y - 3 = 0$ d) $x + y + 3 = 0$

14. The equation of line parallel to $5x + 6y + 1 = 0$ and passing through (2, 3) is

- a) $5x + 6y - 28 = 0$ b) $x + y + 28 = 0$
c) $5x - 6y - 28 = 0$ d) $x - y - 28 = 0$

15. The equation to the line passing through the points (2, 3) and parallel to the line joining the points (-8, -6) and (2, -4) is

- a) $x + 5y + 13 = 0$ b) $5x - y + 13 = 0$
c) $5x - y - 13 = 0$ d) $x - 5y + 13 = 0$

16. The equation of straight line perpendicular to line $x + 5y - 9 = 0$ and passing through the point (-1, 2) is

- a) $5x - y + 14 = 0$ b) $x + 5y - 14 = 0$
c) $5x - y + 7 = 0$ d) $x + 5y + 7 = 0$

17. The equation of the line passing through the point (3, -2) and perpendicular to the line joining the points (5, 2) and (7, -6)

- a) $x - 4y - 11 = 0$ b) $x + 4y - 11 = 0$
c) $x - 4y + 11 = 0$ d) $x + 4y + 11 = 0$

18. The equation of line joining points (2, -3) and (5, 4) is

- a) $7x + 3y - 23 = 0$ b) $7x + 3y + 23 = 0$
c) $3x - 7y + 23 = 0$ d) $7x - 3y - 23 = 0$

19. The equation of line whose y-intercept is 5 units and makes 45° with x-axis is

- a) $x + y + 5 = 0$ b) $x - y + 5 = 0$
c) $x + y - 5 = 0$ d) $x - y - 5 = 0$

20. The equation of straight line having slope $\frac{1}{2}$ and y-intercept 5 is

- a) $2x - y + 10 = 0$ b) $x + 2y + 10 = 0$
c) $x - 2y + 10 = 0$ d) $x - 2y - 10 = 0$

21. The equation of straight line through the origin and inclined at $\frac{\pi}{6}$ radians with x-axis is

- a) $\sqrt{3}x + y - 1 = 0$ b) $x + \sqrt{3}y = 0$
c) $x - \sqrt{3}y = 0$ d) $x - \sqrt{3}y = 0$

22. The equation of line whose angle of inclination is 45° and passing through origin is

- a) $x + y = 0$ b) $x - y = 0$
c) $x + y - c = 0$ d) $x + y + c = 0$

23. Which of the following line of equation pass through origin?

- a) $3x + 2y + 1 = 0$ b) $x - y + 1 = 0$
c) $y = 2x + 4$ d) $x + 2y = 0$

24. The equation of line passing through origin and parallel to $2(x - 1) - 3(y + 4) = 0$ is

- a) $2x + 3y = 0$ b) $3x - 2y = 0$
c) $2x - 3y = 0$ d) $3x + 2y = 0$

25. The slope of the line $2(4x + 3) - 3(2y + 1) = 0$ is

- a) $\frac{5}{3}$ b) $\frac{4}{3}$ c) $\frac{2}{3}$ d) $\frac{7}{3}$

26. The acute angle between the lines $6x + 9y + 15 = 0$ and $12x + 18y + 30 = 0$ is

- a) 0° b) 90° c) 45° d) none

27. The angle between the lines whose slopes are

$\sqrt{3}$ and $\frac{1}{\sqrt{3}}$ is

- a) 45° b) 60° c) 30° d) none

28. The angle between the lines $y = 3$ and $y = \sqrt{3}x + 9$ is

- a) 45° b) 60° c) 30° d) none

29. The angle between the lines $y = x + 4$ and $y = 2x - 3$ is

a) $\tan^{-1}\left(\frac{1}{4}\right)$ b) $\tan^{-1}\left(\frac{1}{3}\right)$

- c) $\tan^{-1}3$ d) none

30. The angle between the lines $y = \sqrt{3}x + 2$ and

$y = \frac{1}{\sqrt{3}}x - 4$ is

- a) $\frac{\pi}{3}$ b) $\frac{\pi}{6}$ c) $\frac{\pi}{4}$ d) $\frac{\pi}{2}$

31. The angle between the lines slopes are $\frac{1}{\sqrt{3}}$ and $\sqrt{3}$ is

- a) 45° b) 60° c) 30° d) 90°

32. The equation of the line which makes intercepts 4 and -2 on the line x-axis and y-axis respectively is

- a) $x - 2y - 2 = 0$ b) $x - 2y - 4 = 0$
c) $4x - 2y - 2 = 0$ d) $x + 2y - 2 = 0$

33. The equation of the line passing through the point (4, 3) and making equal intercepts on the coordinate axis is

- a) $x + y - 7 = 0$ b) $x + 2y - 7 = 0$
c) $x - 2y + 1 = 0$ d) $x + 2y - 2 = 0$

34. The equation of the line which makes intercepts 2 and -3 on the line x-axis and y-axis respectively is

- a) $3x + 2y + 6 = 0$ b) $3x - 2y - 6 = 0$
c) $3x + 2y - 6 = 0$ d) $3x - 2y + 6 = 0$

KEY ANSWERS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (c) | 3. (a) | 4. (a) | 5. (b) | 6. (a) | 7. (d) | 8. (c) | 9. (d) | 10. (b) |
| 11. (c) | 12. (d) | 13. (b) | 14. (a) | 15. (d) | 16. (c) | 17. (a) | 18. (d) | 19. (b) | 20. (c) |
| 21. (c) | 22. (b) | 23. (d) | 24. (c) | 25. (b) | 26. (a) | 27. (c) | 28. (b) | 29. (b) | 30. (b) |
| 31. (c) | 32. (b) | 33. (a) | 34. (b) | | | | | | |

PREVIOUS YEAR MCQ's

- The equation of the straight line passing through (3, 2) and perpendicular to the line $5x + 2y - 3 = 0$ is (DCET - 2013)
 - $2x - 5y - 4 = 0$
 - $2x - 5y + 4 = 0$
 - $2x + 5y + 4 = 0$
 - $5x - 2y + 4 = 0$
- The slope of a line passing through the points (-4, -5) and (2, 3) is (DCET - 2013)
 - $\frac{3}{4}$
 - $-\frac{3}{4}$
 - $\frac{4}{3}$
 - $-\frac{4}{3}$
- The slope of the line perpendicular to the line $2(4x + 3) - 3(2y + 1) = 0$ is, (DCET - 2014)
 - $\frac{4}{3}$
 - $\frac{3}{4}$
 - $-\frac{3}{4}$
 - $-\frac{4}{3}$
- The equation of the line passing through (2, 3) and parallel to the line $5x + 6y + 1 = 0$ is (DCET - 2014)
 - $5x + 6y + 28 = 0$
 - $5x - 6y - 28 = 0$
 - $5x + 6y - 28 = 0$
 - $5x - 6y + 28 = 0$
- The value of k so that the line joining the points (3, 2), (-6, -4) is perpendicular to the line joining the points (k , 4) and (5, 6) is (DCET - 2014)
 - $\frac{19}{3}$
 - $\frac{5}{3}$
 - $-\frac{3}{5}$
 - $-\frac{5}{3}$
- The slope of the line joining the points (-2, 3) and (4, -6) is (DCET = 2015)
 - $\frac{3}{2}$
 - $-\frac{3}{2}$
 - $\frac{2}{3}$
 - $-\frac{2}{3}$
- The equation of straight line passing through (4, -1) and having equal intercepts is (DCET - 2015)
 - $x + y - 1 = 0$
 - $x + y - 5 = 0$
 - $x + y - 3 = 0$
 - $x + y + 3 = 0$
- The equation of the line passing through (5, -2) and parallel to the line $3x + 2y + 7 = 0$ is (DCET - 2015)
 - $3x + 2y - 11 = 0$
 - $3x - 2y + 11 = 0$
 - $3x - 2y + 19 = 0$
 - $2x - 3y + 16 = 0$
- The slope of the line making an angle 150° with the x -axis is (DCET - 2016)
 - $-\frac{1}{\sqrt{3}}$
 - $\frac{1}{\sqrt{3}}$
 - $\sqrt{3}$
 - $-\sqrt{3}$
- The two points form of a straight line is (DCET - 2016)
 - $y - y_1 = m(x - x_1)$
 - $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$
 - $\frac{y}{x} = \frac{y_2 - y_1}{x_2 - x_1}$
 - $\frac{y - y_2}{x - x_2} = \frac{y_2 - y_1}{x_2 - x_1}$
- The equation of straight line perpendicular to $2x + 5y - 8 = 0$ and passing through (-1, 2) is (DCET - 2016)
 - $2x + 5y + 9 = 0$
 - $5x - 2y + 1 = 0$
 - $5x - 2y + 9 = 0$
 - $5x + 2y - 9 = 0$

12. The equation of straight line whose intercepts are 3 and 5 on the axes is (DCET - 2017)

- a) $5x - 3y = 15$ b) $5x + 3y = 15$
c) $5x + 3y = 1$ d) $15x + 15y = 1$

13. The equation of straight line passing through the point $(-6, 7)$ and parallel to the line joining $(3, 4)$ and $(6, -8)$ is (DCET - 2017)

- a) $4x + y + 31 = 0$ b) $x + 4y - 1 = 0$
c) $x - 4y + 1 = 0$ d) $4x + y + 17 = 0$

14. The equation of line passing through the point $(1, -3)$ and having slope $\frac{1}{2}$ is (DCET - 2018)

- a) $x - 2y - 7 = 0$ b) $2x - y + 7 = 0$
c) $x - 2y - 4 = 0$ d) $x + y - 4 = 0$

15. The equation of line passing through the point $(-2, 3)$ and parallel to the line $5x + 3y + 5 = 0$ is, (DCET - 2018)

- a) $5x + 3y - 19 = 0$ b) $5x + 3y + 1 = 0$
c) $5x + 3y + 19 = 0$ d) $3x - 5y + 1 = 0$

16. The slope and y-intercept of the line $6x - 4y + 3 = 0$ are respectively (DCET - 2019)

- a) $\frac{3}{2}$ and $\frac{3}{4}$ b) $\frac{2}{3}$ and $\frac{4}{3}$
c) $-\frac{3}{2}$ and $\frac{4}{3}$ d) $\frac{3}{2}$ and $\frac{2}{3}$

17. The equation of the line joining the points $(1, 3)$ and $(2, -4)$ is (DCET - 2019)

- a) $7x - y - 10 = 0$ b) $7x + y - 10 = 0$
c) $x + 7y + 10 = 0$ d) $x - 7y - 10 = 0$

18. Equation of the straight line passing through the point $(1, 3)$ and having slope -2 is (DCET - 2020)

- a) $2x - y + 5 = 0$ b) $x + 2y + 5 = 0$
c) $x - 2y - 5 = 0$ d) $2x + y - 5 = 0$

19. Equation of the straight line passing through the origin and perpendicular to the line $5x - 4y - 1 = 0$ is (DCET - 2020)

- a) $5x - 4y = 0$ b) $4x + 5y = 0$
c) $5x - 4y + 1 = 0$ d) $4x + 5y + 1 = 0$

20. The equation of the straight line passing through $(2, 3)$ and parallel to $2x + 3y + 6 = 0$ is _____. (DCET - 2021)

- a) $2x + 3y - 13 = 0$ b) $3x + 2y + 13 = 0$
c) $3x + 2y - 13 = 0$ d) $2x + 3y + 13 = 0$

21. The slope of the line perpendicular to the line $3x - 2y + 1 = 0$ is _____. (DCET - 2021)

- a) $\frac{2}{3}$ b) $\frac{3}{2}$ c) $-\frac{2}{3}$ d) $-\frac{3}{2}$

22. The x-intercept and y-intercept of the line $3x - 4y + 1 = 0$ are (DCET - 2022)

- a) $\frac{1}{3}$ and $\frac{1}{4}$ b) $-\frac{1}{3}$ and $-\frac{1}{4}$
c) $-\frac{1}{3}$ and $\frac{1}{4}$ d) $\frac{1}{3}$ and $-\frac{1}{4}$

23. The equation of the line passing through the points $(1, -2)$ and $(2, 1)$ is (DCET - 2022)

- a) $3x + y + 5 = 0$ b) $3x - y + 5 = 0$
c) $3x - y - 5 = 0$ d) $x + 3y + 5 = 0$

24. The angle of inclination of the line $x + y - 2 = 0$ is (DCET - 2022)

- a) $-\frac{\pi}{4}$ b) $\frac{\pi}{4}$
c) $\frac{3\pi}{4}$ d) $-\frac{3\pi}{4}$

KEY ANSWERS

1. (b) 2. (c) 3. (c) 4. (c) 5. (a) 6. (b) 7. (c) 8. (a) 9. (a) 10. (b)
11. (c) 12. (b) 13. (d) 14. (a) 15. (b) 16. (a) 17. (b) 18. (d) 19. (b) 20. (a)
21. (c) 22. (c) 23. (c) 24. (c)